Unified Glance-Focus pretraining and task adaptation

Heterogeneous modalities share one representation; incompatible labels remain in separate task heads

1 PRETRAINING TASKS

Eight original task schemas

Brain MRI ± CT/RTP
BrainMets · MU-Glioma · UCSD-PTGB · Glioma
T1 variants · T2 variants · FLAIR · ADC · CT · RTP

Prostate MRI
PI-CAI · T2W + ADC + HBV

Chest CT
LUNA25 · lung-nodule CT

Whole-body PET/CT
PETWB Lung · PETWB Liver
paired CT + FDG PET

9,945 cases · 1-6 named channels per examination

2 TASK SAMPLING

Sample one task

One dataset is selected for each training step

Task-homogeneous batch
images + valid labels

Known task key
modality IDs + head schema

3 SHARED BACKBONE

Shared modality-aware Glance-Focus

Modality-set stem
shared channel encoder + modality embedding + attention fusion

GLANCE
whole-volume context
global feature

FOCUS
local encoder-decoder
mask + lesion feature

Shared classification trunk + task embedding
common features, task-conditioned interpretation

4 TASK-SPECIFIC SUPERVISION

Task-specific head bank

The known task key activates one compatible head set

Brain tasks
BrainMets · MU-Glioma · UCSD-PTGB · Glioma
separate segmentation + clinical heads for each dataset

PI-CAI
prostate/tumor + ISUP

LUNA25
nodule + malignancy

PETWB Lung
lung + cancer status

PETWB Liver
liver + cancer status

Only the selected heads enter the loss
inactive heads produce no logits and receive no gradient

Unified pretrained checkpoint

shared representation learned from all eight tasks; label spaces remain separate

Later adaptation
add new EGCT or PHLF heads
load shared weights, then fine-tune task-only

ONE SHARED BACKBONE | ONE TASK PER TRAINING STEP | ONE TASK-SPECIFIC HEAD SET ACTIVE | EGCT AND PHLF ARE LATER FINE-TUNING TASKS